
Yun Qiao

Columbus, OH | (614) 981 8270 | qiao.247@osu.edu | <https://github.com/Yun-Qiao11283>

EDUCATION

The Ohio State University

Columbus, OH

Bachelor of Science in Mathematics, Minor in Computer Science

Expected May 2027

- **GPA: 3.981 / 4.0** — Honors Program Student
- **Honors & Awards:**
 - * **Dean's List:** Recognized for 5 consecutive semesters (AU23, SP24, AU24, SP25, AU25), representing the top 10% of the department.
 - * **Mathematics Department Scholarship:** Awarded for outstanding academic performance and potential in mathematical research.
 - * **First Prize for Academic Scholarship:** Received for maintaining a near-perfect GPA in rigorous Honors-level coursework.
- **Core Course**
 - * **Operations Research & Statistics**
 - Linear Programming (ISE 5200) ***Grad Level***
 - Mathematical Statistics (STAT 4201/4202)
 - Machine Learning (MATH 5638) ***Grad Level***
 - * **Computer Science**
 - Data Structures in C++ (CSE 2122)
 - Discrete Structures (CSE 2321)
 - Programming in C++ (CSE 1222)
 - * **Advanced Math**
 - Honors Real Analysis (4181H/4182H) ***Grad Level***
 - Honors Linear Algebra & Differential Equation (5520H) ***Grad Level***
 - Honors Complex Analysis (5522H) ***Grad Level***
 - General Topology & Knots (5801) ***Grad Level***

RESEARCH INTERESTS

- **Large-Scale Computational Optimization:** Developing robust second-order and quasi-Newton algorithms to overcome convergence bottlenecks and minimize the computational overhead of sparse matrix operations in high-dimensional machine learning.
- **Numerical Linear Algebra & Iterative Solvers:** Designing advanced preconditioning strategies and Krylov subspace methods for ill-conditioned sparse systems to accelerate PDE solvers and reduce Hessian inversion complexity in large-scale optimization.

RESEARCH EXPERIENCE

Research Assistant

Under Supervision of Prof. Anna Yesypenko, The Ohio State University

Jan 2026 – Present

This research develops high-performance numerical solvers for PDEs where analytical solutions are unavailable. We implement discretization schemes (FDM/FEM) to transform continuous operators into algebraic systems, utilizing iterative methods—specifically Multigrid and Preconditioned Conjugate Gradient—to solve large-scale sparse matrices. The objective is to optimize the trade-off between discretization accuracy and computational efficiency for high-fidelity simulations in fluid and structural mechanics.

Topological Data Analysis (TDA) for Equity Markets

Under Supervision of Ph.D. Candidate Krnic Luci, The Ohio State University

Sep 2025 – Present

Investigated the limitations of traditional financial models in capturing high-dimensional structural shifts during macroeconomic shocks by applying Topological Data Analysis (TDA), specifically persistent homology, to a comprehensive market manifold comprising core US equities and exogenous safe-haven assets. Currently developing a Python-based computational pipeline to construct simplicial complexes and extract the L_1 norm of Persistence Landscapes to rigorously quantify topological phase transitions, ultimately delivering a robust, threshold-free early warning system to anticipate systemic market crashes.